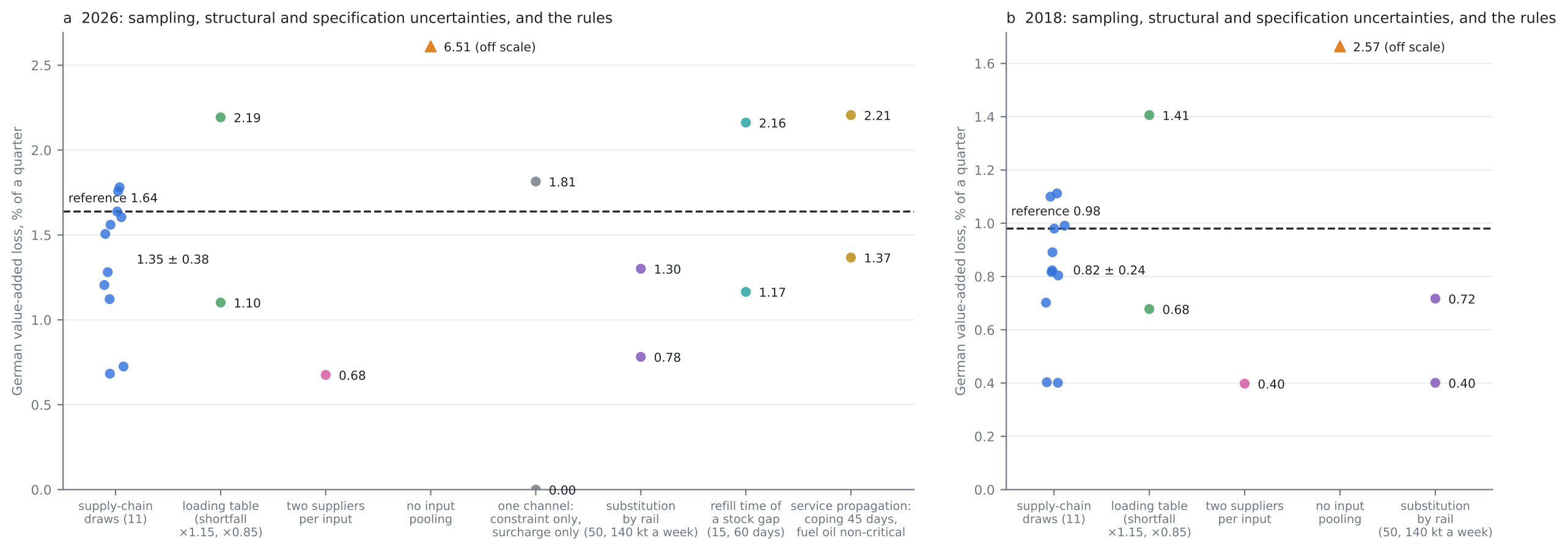
a 2026: sampling, structural and specification uncertainties, and the rules
6.51 (off scale)
German value-added loss, % of a quarter
2.5
2.0
1.5
1.0
0.5
0.0
2.19
2.16
2.21
1.81
reference 1.64
1.35 ± 0.38
1.30
1.37
1.17
1.10
0.78
0.68
0.00
supply-chain draws (11)
loading table (shortfall ×1.15, ×0.85)
two suppliers per input
no input pooling
one channel: constraint only, surcharge only
substitution by rail (50, 140 kt a week)
refill time of a stock gap (15, 60 days)
service propagation: coping 45 days, fuel oil non-critical
b 2018: sampling, structural and specification uncertainties, and the rules
2.57 (off scale)
German value-added loss, % of a quarter
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.41
reference 0.98
0.82 ± 0.24
0.68
0.72
0.40
0.40
supply-chain draws (11)
loading table (shortfall ×1.15, ×0.85)
two suppliers per input
no input pooling
substitution by rail (50, 140 kt a week)